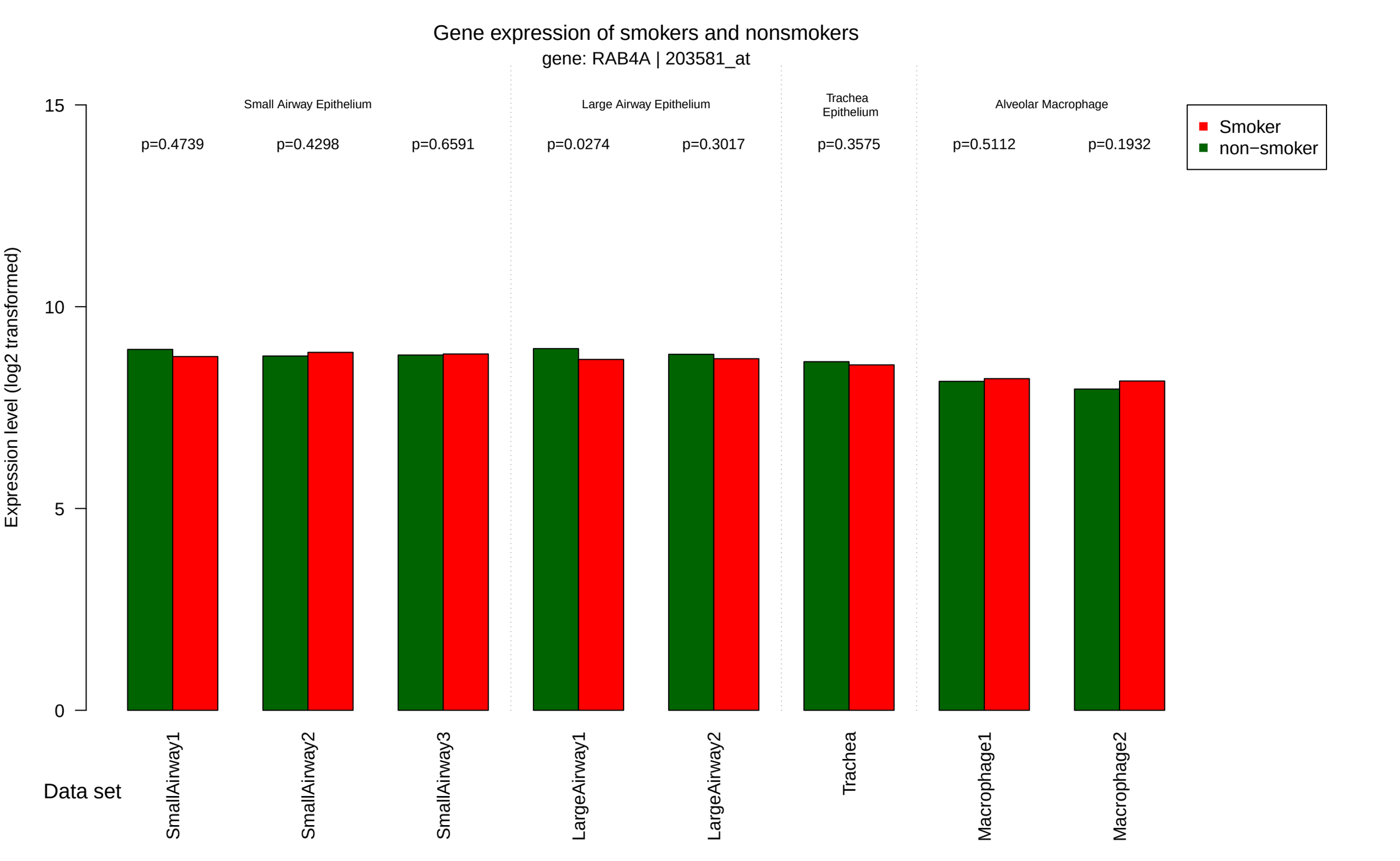
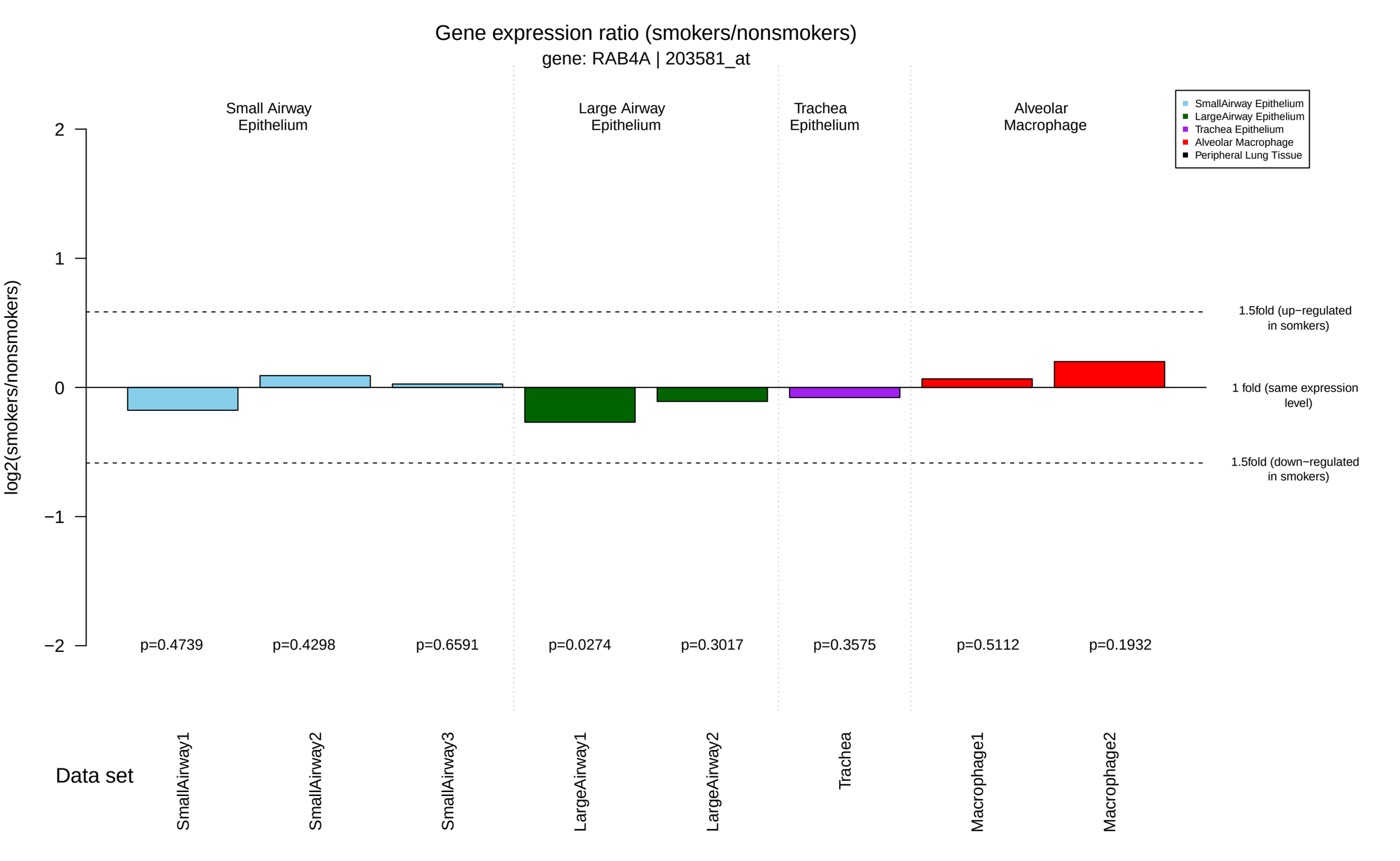
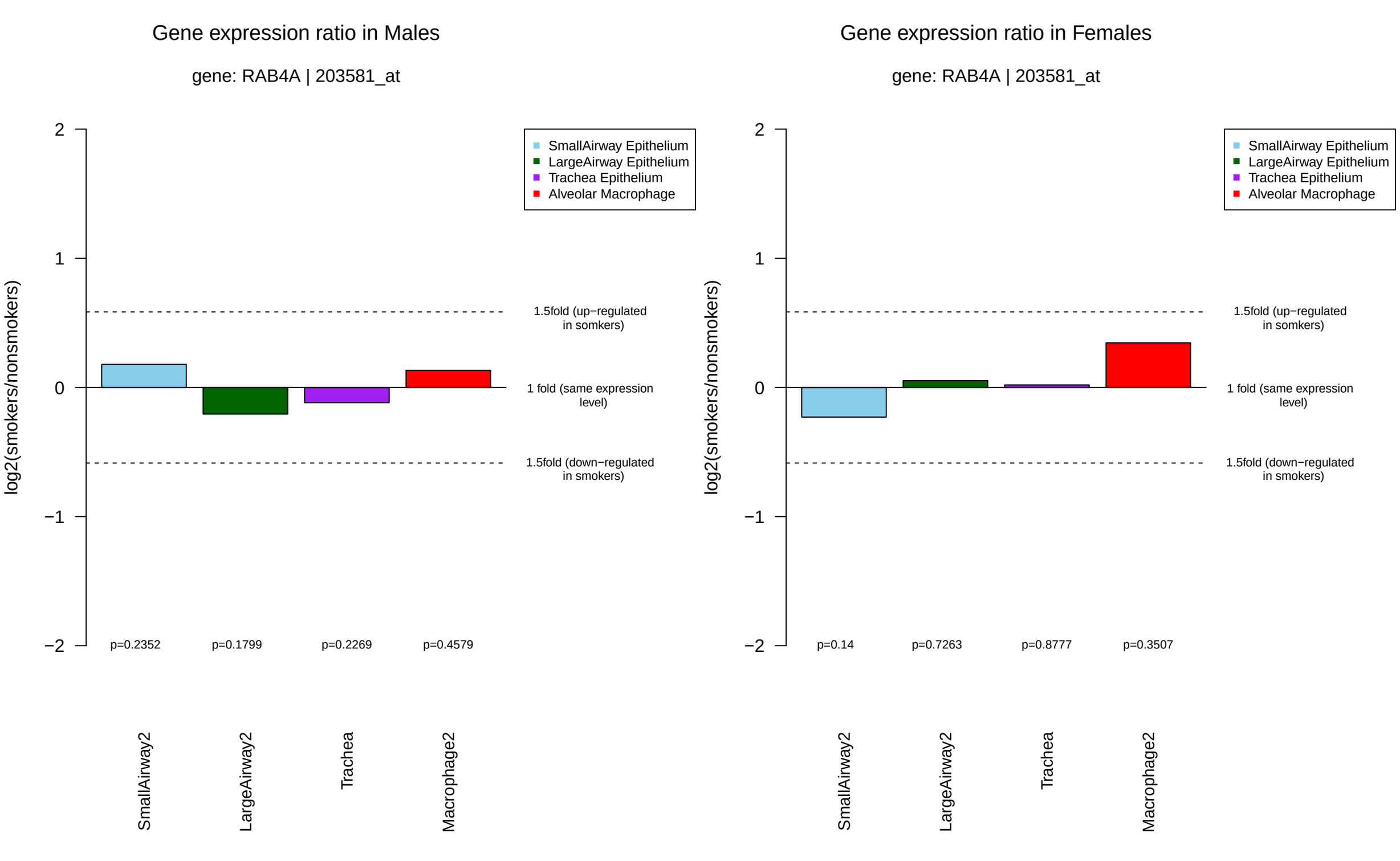
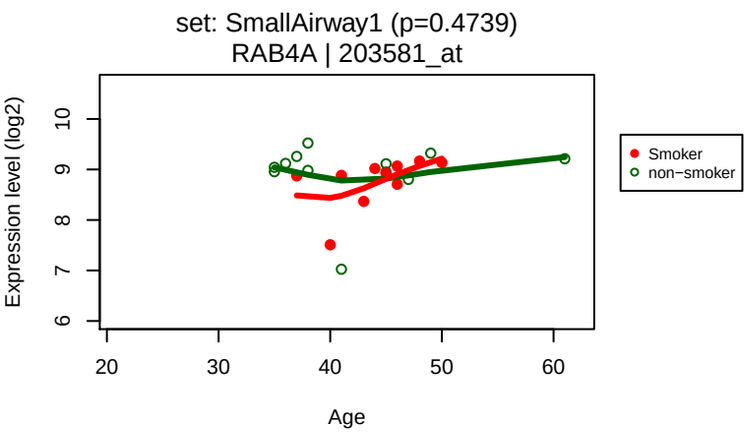
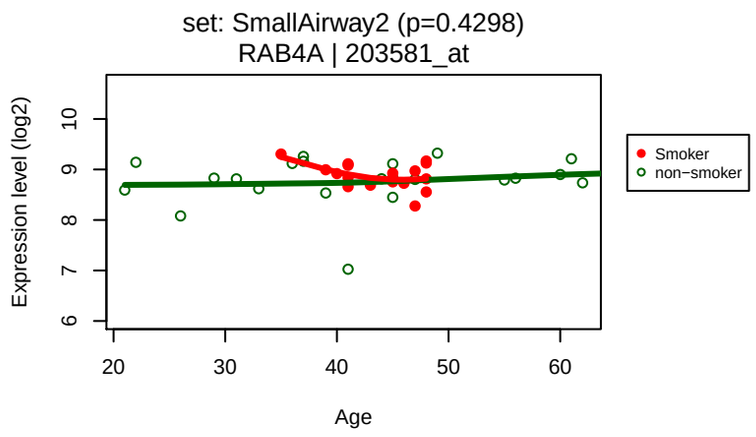
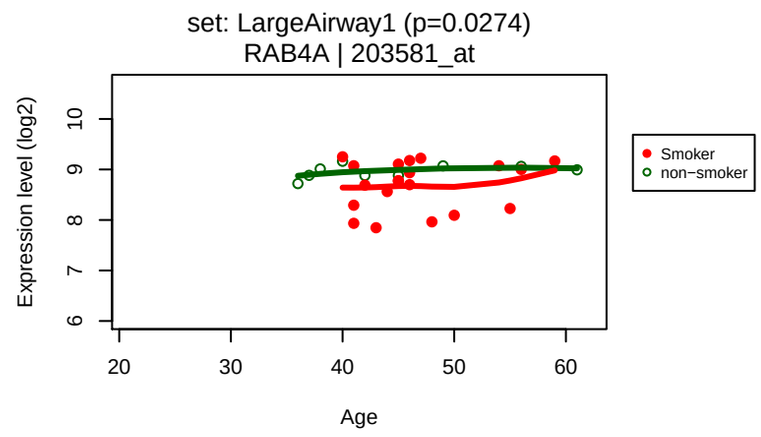

Gene expression of smokers and nonsmokers
gene: RAB4A | 203581_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4739
p=0.4298
p=0.6591
p=0.0274
p=0.3017
p=0.3575
p=0.5112
p=0.1932
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

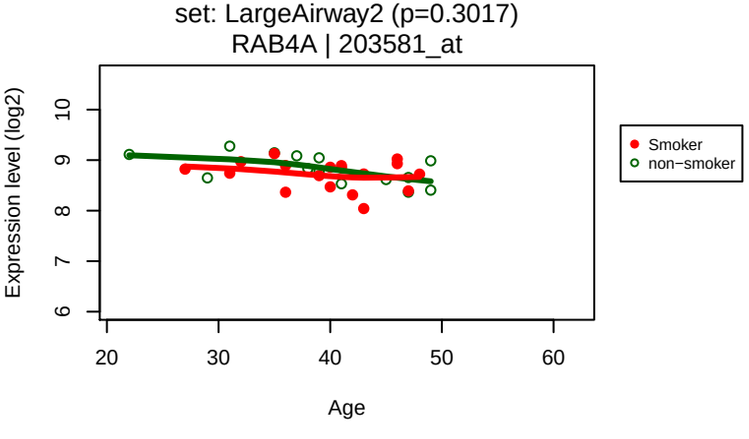

Gene expression ratio (smokers/nonsmokers)
gene: RAB4A | 203581_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4739
p=0.4298
p=0.6591
p=0.0274
p=0.3017
p=0.3575
p=0.5112
p=0.1932
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

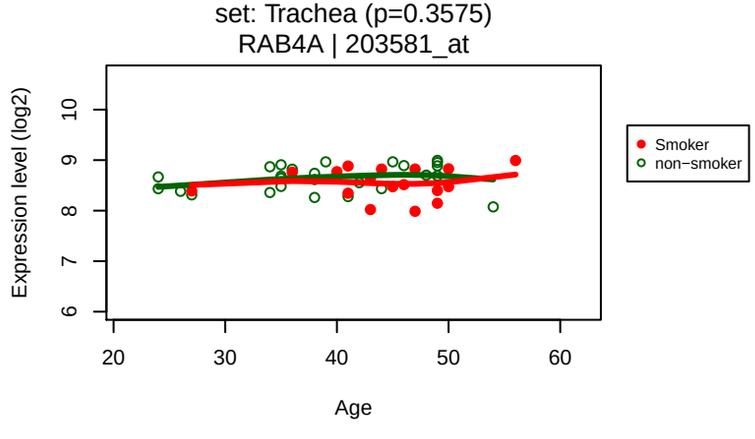

Gene expression ratio in Males
gene: RAB4A | 203581_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2352
p=0.1799
p=0.2269
p=0.4579
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB4A | 203581_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.14
p=0.7263
p=0.8777
p=0.3507
SmallAirway2
LargeAirway2
Trachea
Macrophage2

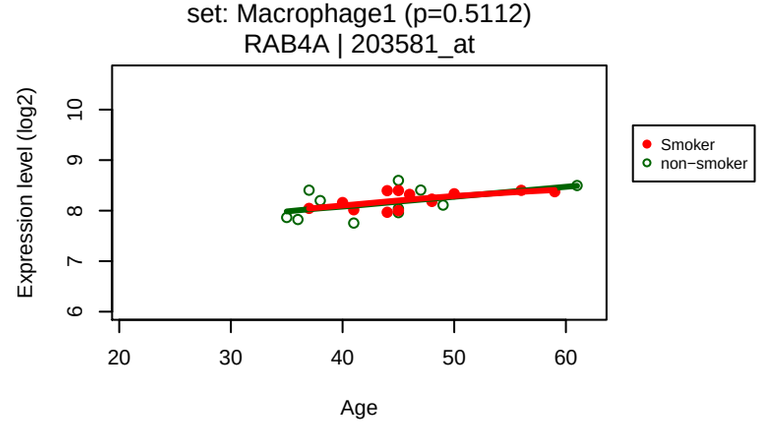

set: SmallAirway1 (p=0.4739)
RAB4A | 203581_at
Expression level (log2)
Age
Smoker
non−smoker

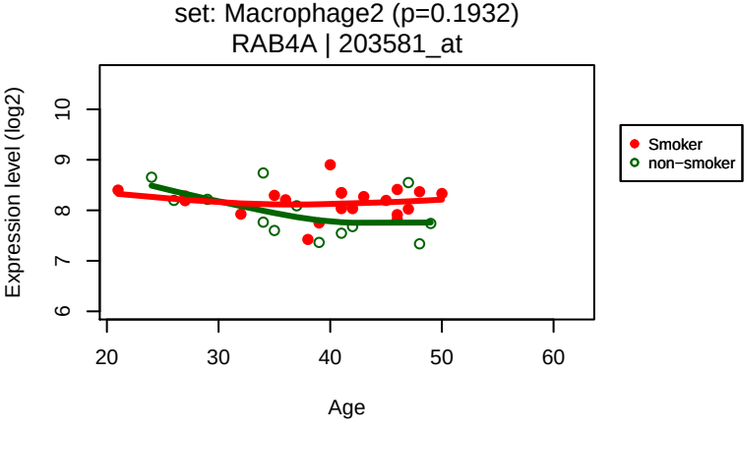

set: SmallAirway2 (p=0.4298)
RAB4A | 203581_at
Expression level (log2)
Age
Smoker
non−smoker

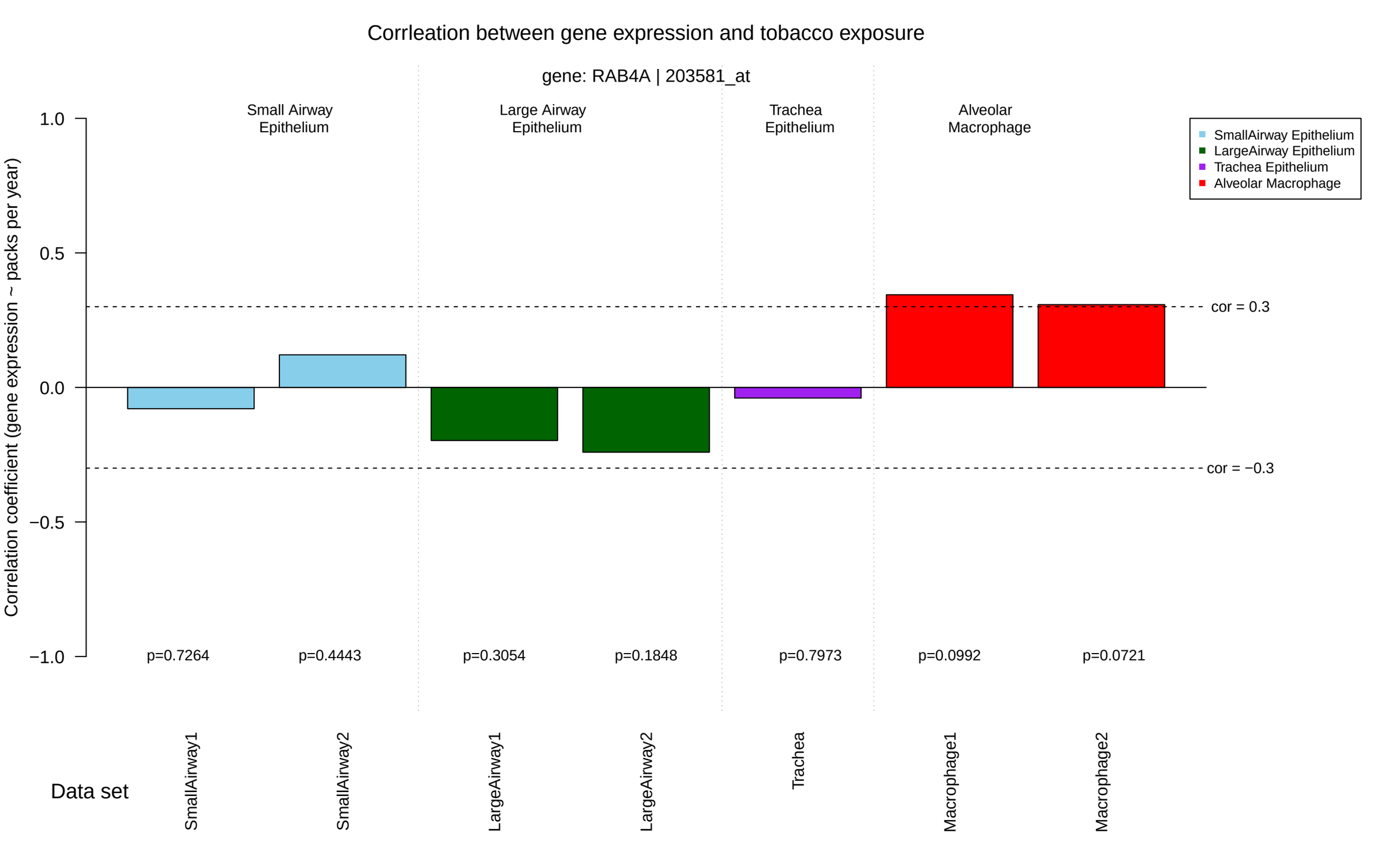

set: LargeAirway1 (p=0.0274)
RAB4A | 203581_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3017)
RAB4A | 203581_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3575)
RAB4A | 203581_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5112)
RAB4A | 203581_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1932)
RAB4A | 203581_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB4A | 203581_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7264
p=0.4443
p=0.3054
p=0.1848
p=0.7973
p=0.0992
p=0.0721
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2